

ORAN, Argentina

WMO: 870160

Lat: 23.1547S

Lon: 64.3281W

Elev: 1171

StdP: 14.08

Time Zone: -3.00 (W03)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.8	44.9	34.1	30.0	57.5	37.3	34.2	54.7	15.7	61.3	12.0	61.5	3.0	0	0.289

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	17.4	98.6	75.3	95.9	75.4	93.4	75.5	80.6	90.6	79.6	89.3	78.6	88.0	6.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.2	153.3	85.7	77.1	147.3	84.7	76.1	142.4	83.7	45.5	90.3	44.3	89.5	43.3	88.1	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.7	12.1	10.2	DB	39.1	105.1	3.8	3.0	36.3	107.3	34.1	109.0	31.9	110.7	29.2	112.9
			WB	37.8	83.4	3.5	2.0	35.3	84.8	33.2	86.0	31.3	87.1	28.7	88.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	72.2	80.5	79.3	75.8	71.9	65.9	61.0	60.3	66.0	70.4	76.6	79.1	80.5
	DBStd	9.39	4.55	4.76	5.26	5.89	6.08	6.03	6.17	7.47	7.20	7.08	5.83	5.36
	HDD50	6	0	0	0	0	0	2	4	0	0	0	0	0
	HDD65	511	0	0	2	14	62	149	172	81	27	4	0	0
	CDD50	8124	946	819	800	658	495	331	322	497	611	825	873	947
	CDD65	3150	481	399	337	221	91	28	25	112	188	363	423	482
	CDH74	26884	4565	3483	2279	1113	294	80	150	857	1571	3479	4180	4833
	CDH80	11219	1920	1388	742	266	41	5	21	296	634	1662	1987	2257
Wind	WSAvg	3.8	4.2	4.1	3.5	3.1	2.7	2.8	3.2	3.7	4.2	4.6	4.6	4.3
Precipitation	PrecAvg	37.6	7.8	6.5	6.0	3.1	0.7	0.4	0.2	0.2	0.4	2.0	3.8	6.1
	PrecMax	54.1	18.1	14.7	13.5	7.9	3.0	1.3	0.9	1.1	1.6	6.7	9.5	11.9
	PrecMin	25.8	1.4	1.9	1.6	0.3	0.0	0.0	0.0	0.0	0.0	0.2	0.2	2.0
	PrecStd	8.2	3.9	3.6	3.0	2.0	0.7	0.4	0.2	0.3	0.4	1.5	2.1	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.8	97.4	93.9	89.1	84.0	80.6	83.5	92.2	96.5	102.1	101.6	101.7
		MCWB	78.6	77.2	78.6	76.0	71.9	68.3	67.9	70.3	70.6	74.4	74.0	76.5
	2%	DB	94.4	93.5	90.3	85.9	80.3	76.5	78.6	87.4	91.6	97.2	97.7	97.5
		MCWB	78.7	78.1	77.4	75.1	70.9	66.4	65.7	69.1	69.5	73.9	74.4	76.9
	5%	DB	92.1	90.7	87.4	83.0	77.2	73.1	74.7	82.6	87.4	93.4	94.1	94.0
		MCWB	78.2	77.4	77.0	74.0	69.6	64.9	62.9	66.8	67.8	72.3	73.7	76.5
	10%	DB	89.5	88.1	84.3	80.1	74.3	70.0	71.1	78.4	83.3	89.6	90.6	90.8
		MCWB	77.4	76.9	75.4	72.9	68.4	63.2	61.2	64.0	66.0	71.1	72.9	76.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.5	82.0	80.4	79.0	74.7	70.3	69.4	73.0	73.8	77.3	79.9	81.0
		MCDB	90.8	90.5	88.8	86.3	80.4	77.7	80.4	87.0	89.8	94.8	93.9	92.0
	2%	WB	80.8	80.1	79.0	77.2	72.7	68.1	66.9	70.3	71.7	75.7	77.8	79.4
		MCDB	90.4	88.8	87.3	83.3	78.0	73.6	76.3	84.3	87.0	92.8	90.3	91.2
	5%	WB	79.7	78.9	77.7	75.8	71.0	66.4	64.5	68.0	69.7	74.1	76.2	78.2
		MCDB	89.1	87.5	85.1	80.7	75.7	71.2	72.6	79.0	82.9	89.3	87.9	89.7
	10%	WB	78.4	77.7	76.5	74.3	69.2	64.6	62.6	65.9	67.5	72.6	74.8	77.1
		MCDB	87.2	85.9	82.9	78.4	73.0	69.2	68.7	76.6	80.6	85.9	85.6	88.0
Mean Daily Temperature Range	5% DB	MDBR	16.1	15.7	13.7	12.7	12.6	15.4	18.9	21.6	21.1	19.8	18.8	17.4
		MCDBR	20.2	19.6	18.0	17.1	16.2	20.4	24.9	27.2	28.0	26.7	25.6	23.3
		MCWBR	7.1	7.0	7.0	7.0	7.0	10.5	11.8	10.5	9.8	8.5	7.9	7.5
	5% WB	MCDBR	18.2	17.9	16.0	14.2	13.7	16.4	22.1	24.7	24.4	23.7	20.8	20.4
		MCWBR	7.9	7.6	7.0	7.0	6.4	9.1	10.9	10.0	9.9	8.1	8.2	8.0
Clear-Sky Solar Irradiance	taub	0.426	0.424	0.417	0.400	0.364	0.331	0.334	0.398	0.487	0.494	0.441	0.435	
	taud	2.355	2.371	2.392	2.406	2.435	2.506	2.474	2.276	2.036	2.084	2.268	2.317	
	Ebn at Noon	292	288	281	272	270	274	278	268	257	265	286	290	
	Edh at Noon	43	41	38	35	31	28	30	39	54	54	46	44	
All-Sky Solar Radiation	RadAvg	1891	1738	1438	1194	1022	992	1165	1432	1627	1679	1820	1826	
	RadStd	156	111	134	124	111	88	89	113	148	146	177	129	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-25	N/A	N/A	N/A

Nomenclature: See separate page